

Decimal Multiplication & Division Workshop

Answer Key

Grade 5

Quick Answers

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|---|---|
| 1. 360 | 7. 9 |
| 2. 4.52 | 8. 240 kg |
| 3. 0.24 | 9. (a) each friend's share = 2.45, (b) comparison with 2.50 = < |
| 4. (a) estimate = 8, (b) exact cost = 9.4 | 10. (a) cost of the milk = 3, (b) change from 5 dollars = 2 |
| 5. 8 | |
| 6. 15 L | |
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Curriculum

Level: Grade 5

5.NBT.A.2 – problems 1, 2, 8

5.NBT.B.7 – problems 3, 4, 5, 6, 7, 9, 10

Difficulty 1–4 of 5 across 13 tagged checks.

Common wrong answers

3. *If they got 2.4: counted only one decimal place in the product (24 tenths instead of 24 hundredths)*
5. *If they got 0.8: moved the decimal point in only one of the two numbers*
8. *If they got 24: multiplied by 10 instead of 100*
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1. Multiply: 3.6×100

Solution: Multiplying by 100 makes every digit worth 100 times more, so the decimal point moves *two* places to the right: $3.6 \rightarrow 36 \rightarrow 360$. 360

2. Divide: $45.2 \div 10$

Solution: Dividing by 10 makes every digit worth ten times less, so the decimal point moves *one* place to the left: $45.2 \rightarrow 4.52$. 4.52

3. Multiply: 0.4×0.6

Solution: Multiply as whole numbers: $4 \times 6 = 24$. The factors have one decimal place each, so the product needs *two* decimal places: tenths $\times$ tenths = hundredths, giving 24 hundredths. 0.24

4. Notebooks: 4 at 2.35 dollars each

Solution: (a) Round the price to the nearest whole dollar: $2.35 \rightarrow 2$. Then estimate 4×2 , giving about 8 dollars.

(b) Exact: $4 \times 2.35 = 4 \times 2 + 4 \times 0.35 = 8 + 1.40$, so the total is 9.40 dollars. The estimate of 8 tells us 9.40 is reasonable.

5. Divide: $7.2 \div 0.9$

Solution: Multiply *both* numbers by 10 so the divisor becomes a whole number: $7.2 \div 0.9 = 72 \div 9$. Both numbers shifted together, so the quotient is unchanged. $72 \div 9 = 8$. 8

6. Watering can: 2.5 L filled 6 times

Solution: Total = 2.5×6 . Think money: 2.5 is 2 and 5 tenths, so 6 groups make $12 + 3.0 = 15.0$. 15 L

7. Ribbon: 4.5 m cut into 0.5-m pieces

Solution: “How many 0.5s fit in 4.5?” Multiply both by 10: $45 \div 5 = 9$. So there are 9 pieces.

8. Copy paper: 2.4 kg per box, 100 boxes

Solution: 2.4×100 moves the decimal point two places to the right: $2.4 \rightarrow 24 \rightarrow 240$. 240 kg

9. Trading cards: 7.35 dollars shared by three friends

Solution: (a) $7.35 \div 3$: share the wholes ($7 \div 3 = 2$ with 1 left), trade the leftover dollar for tenths: 13 tenths $\div 3 = 4$ tenths with 1 tenth left, trade for hundredths: $15 \div 3 = 5$. Each friend pays 2.45 dollars.

(b) Compare place by place: 2.45 has 4 tenths, 2.50 has 5 tenths, so $2.45 < 2.50$

10. Challenge: milk at 1.2 dollars per liter, 2.5 liters, 5-dollar bill

Solution: (a) Cost = 1.2×2.5 . As whole numbers $12 \times 25 = 300$; the factors have two decimal places between them, so the cost is 3.00 dollars. 3

(b) Change = $5 - 3.00$, so Priya gets 2 dollars back.